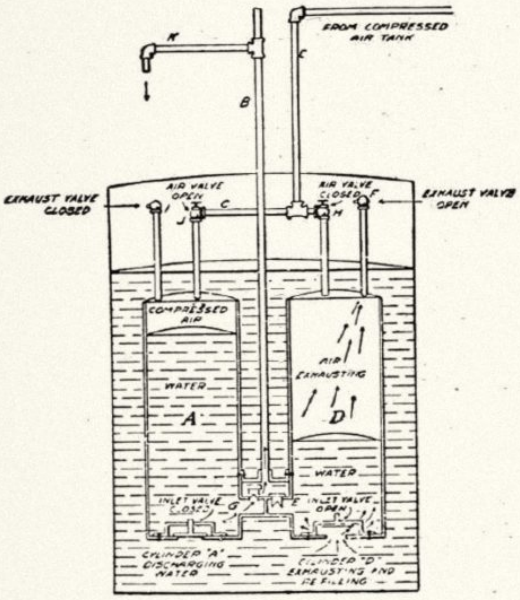
The Autopneumatic System.—In the autopneumatic system the water is delivered fresh from the well to the faucets. This system consists essentially of an air compressor driven by a small gas engine or electric motor, an air-tight steel air-pressure tank, and one or more autopneumatic pumps. No water tank is required, since nothing is stored but compressed air. The pump consists of two small metallic chambers submerged in the water.

When a faucet is opened they automatically fill and discharge, owing to the air pressure from the storage tank. This gives a continuous flow of fresh water.



Fic. 16.—Principle of operation of an autopneumatic pump.

Figure 16 illustrates the principle of operation of the pump. Suppose a small air-tight tank A with inlet valve is submerged in water and allowed to fill. A discharge pipe B is connected at the bottom leading upward to faucet K. Compressed air is forced through pipe C into the top of the tank so that the water is forced out through the discharge pipe until the tank is emptied of water. A similar tank D is connected as shown. The pump has a device for automatically opening and closing the air valves and exhaust valves to the tanks alternately. While A is emptying of water D is filling, and they discharge alternately.

Figure 17A gives a front and side view of an autopneumatic pump. Figure 17B shows the entire working parts of a system and also how the pump may be used in bored and cased wells.

Each pump requires an air-pressure reducer, shut-off and release cocks, pressure gauge, etc. The air-pressure reducer is necessary to reduce the high pressure

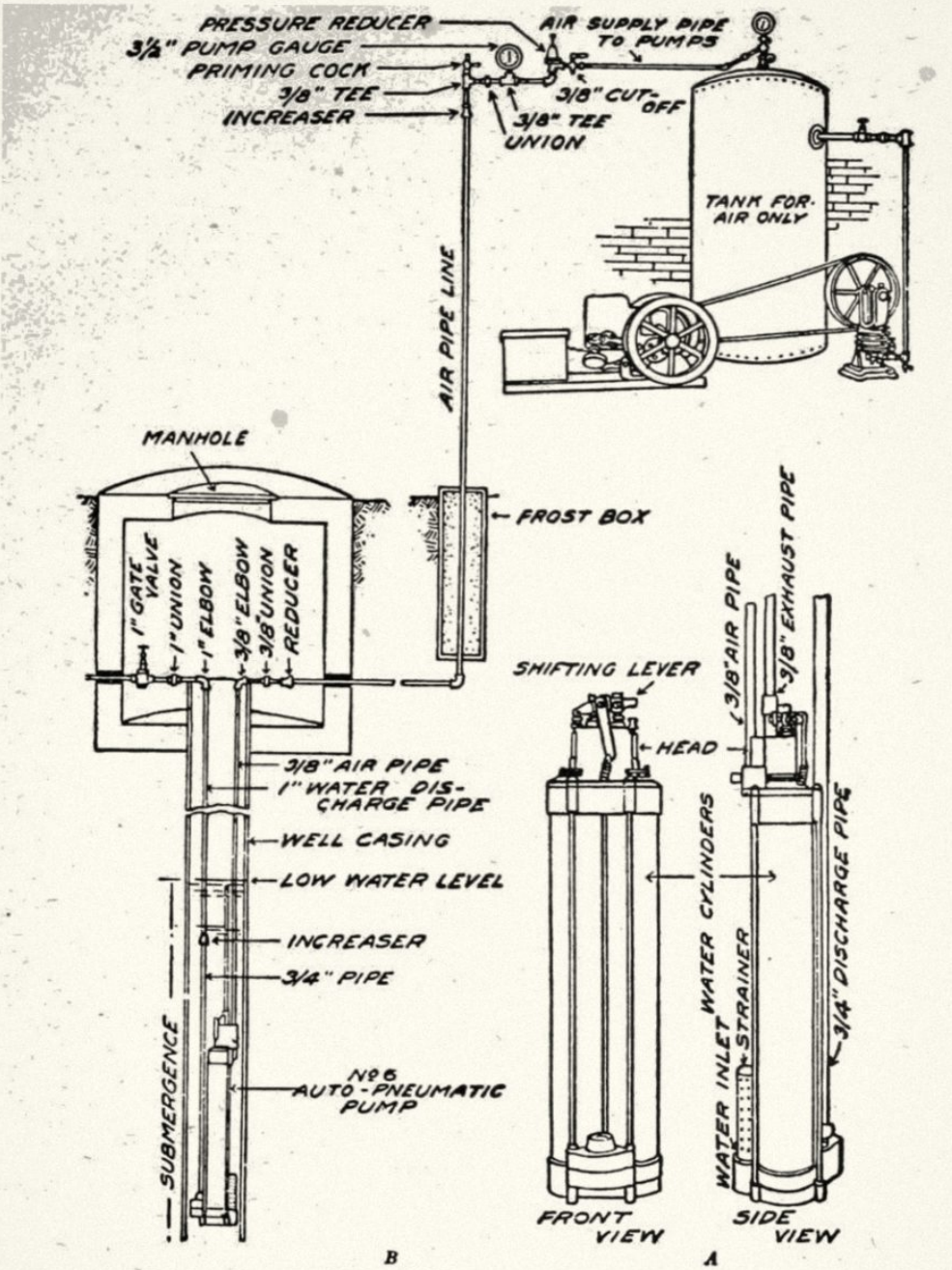
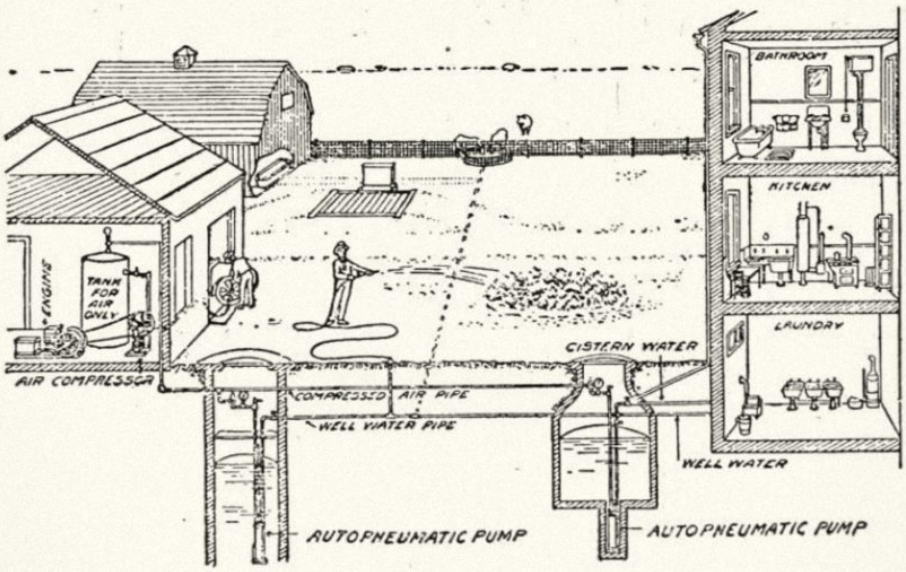


FIG. 17.—Working parts of an autopneumatic pumping system: *A*, Front and side view; *B*, entire working system as used in bored or cased wells.

carried in the tank to the uniform low pressure required to operate the pump. It is placed in the air-pipe line between the air tank and the pneumatic pump. It can be adjusted to the proper pressure with an ordinary wrench. The proper working pressure required to operate the pump and raise water to the required height in the building is recorded on the pump gauge placed on the air-pipe line between the reducer and the pump. The following table shows the number of gallons of water that can be drawn from faucets with the pump under working pressures varying from 25 to 65 pounds and total starting pressures in a 1,000-gallon air tank varying from 40 to 100 pounds:

Pumping copacity of a 1,000-gailon air tank, in gallons, under varying internal pressures.

Work pressure on pump guage.	Total pressure in air tank at start.						
	40 pounds.	50 pounds.	60 pounds.	70 pounds.	80 pounds.	90 pounds.	100 pounds.
Pounds.	*Gallons.*	*Gallons.*	*Gallons.*	*Gallons.*	*Gallons.*	*Gallons.*	*Gallons.*
25	375	259	833	1,075	1,310	1,548	1,786
30	221	442	663	884	1,105	1,326	1,548
35	102	306	510	714	924	1,123	1,327
40	.....	187	374	561	748	936	1,123
45	.....	85	255	425	596	765	936
50	.....	.....	153	306	460	612	765
55	.....	.....	68	204	330	476	612
60	.....	.....	.....	119	237	375	476
65	.....	.....	.....	51	153	255	357



Fic. 18.—A pplication of autopnevmatic sysicm to a farm home.

For air tanks of other than 1,000-gallon capacity divide the figures in the table by 1,000, move the decimal point three places to the left, and multiply the result by the capacity of the tank in gallons.

Assume, for example, that the height to which the water is to be raised and the pressure required to operate the pump, including friction, make necessary a

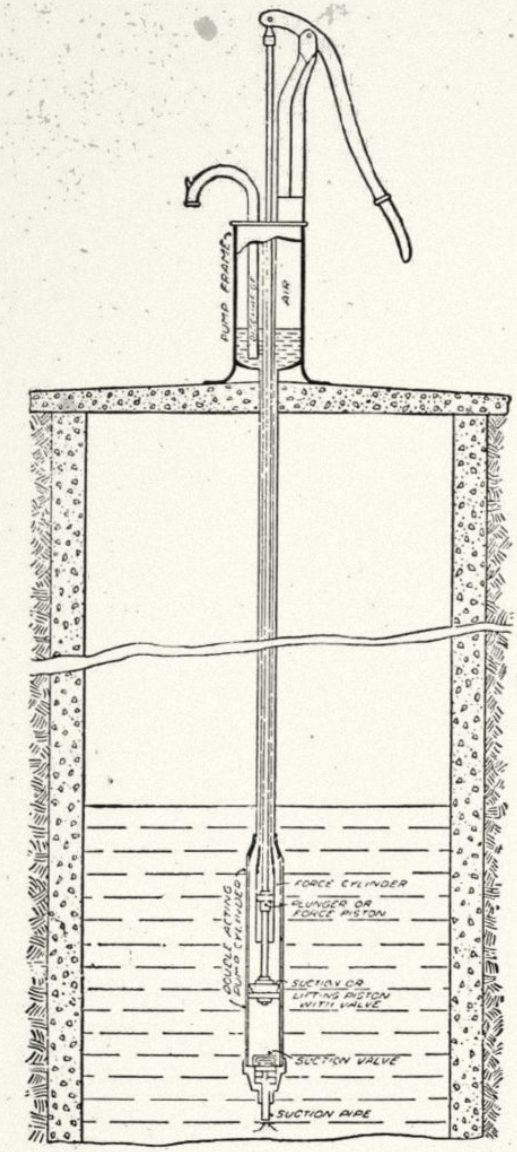


FIG. 19.—Force pump with cylinder submerged in shallow well.

working pressure of 40 pounds on the pump. Then if the initial pressure in the air tank is 50 pounds, 187 gallons of water can be delivered at the faucets before the pressure in the air tank becomes too low to operate the pump.

The autopneumatic pump can be used in wells, springs, or lakes where the water is free from sand and mud and does not have to be lifted more than 100 feet, or where the working pressure on the pump does not exceed 65 pounds.

Figure 18 illustrates the application of this system to a farm home.

Pumping.—The water level in shallow wells is usually near enough to the surface to be within the limits of suction. The limiting practical suction lift for a pump is about 20 feet, although it will vary with the elevation above sea level. This means that the pump cylinder which raises the water by suction in lift pumps and which raises by suction and also forces the water in force pumps should not be more than 20 feet above the water level in the well. To practically eliminate suction lift, the cylinder may be submerged as shown in figure 19, thus making the cylinder and pump frame separate and connected only by a section of pipe. This prevents the valves from drying out and makes the pump self-priming. Force pumps often have two cylinders, and in deep wells it is necessary that the lower or suction cylinder be either submerged or within 15 feet of the water level. In the figure shown the forcing cylinder is within the suction cylinder. The suction of any type of pump must be air-tight.

To find the approximate discharge at each stroke of a pump in gallons use the following table:

TABLE OF CAPACITY OF PUMPS

Diameter of cylinder in inches	Length of stroke in inches									
	5	6	7	8	9	10	12	14	15	16
	Capacity per stroke in gallons									
1	0.017	0.020	0.024	0.027	0.031	0.034	0.041	0.048	0.051	0.054
1⅛	.022	.026	.030	.034	.039	.043	.052	.060	0.65	.069
1¼	.027	.032	0.37	.043	.048	.053	.064	.074	.079	.085
1⅜	.032	.039	.044	.051	.058	.064	.077	.089	.096	.103
1½	.038	.046	.054	.061	.069	.077	.092	.107	.115	.122
1¾	.052	.063	.073	.083	.094	.104	.125	.146	.156	.170
2	.068	.082	.095	.109	.122	.136	.163	.190	.204	.218
2¼	.086	.103	.121	.138	.155	.172	.206	.241	.258	.275
2½	.106	.128	.149	.170	.191	.213	.255	.298	.319	.340
2¾	.129	.154	.180	.206	.231	.257	.309	.360	.386	.411
3	.153	.184	.214	.245	.275	.306	.367	.428	.459	.489
3¼	.179	.215	.251	.287	.323	.359	.431	.503	.539	.575
3½	.208	.249	.292	.333	.375	.417	.499	.583	.625	.666
3¾	.239	.287	.335	.382	.430	.478	.574	.669	.717	.765
4	.272	.326	.381	.435	.490	.544	.653	.762	.816	.870
4¼	.307	.368	.429	.491	.553	.614	.737	.860	.921	.982
4½	.344	.413	.482	.551	.619	.689	.826	.964	1.033	1.102
4¾	.384	.460	.537	.614	.690	.767	.920	1.073	1.150	1.227
5	.425	.510	.595	.680	.765	.850	1.020	1.190	1.275	1.360
5¼	.469	.562	.656	.750	.843	.937	1.124	1.311	1.405	1.499
5½	.514	.617	.720	.823	.926	1.029	1.234	1.440	1.543	1.646
5¾	.562	.674	.787	.899	1.011	1.124	1.348	1.573	1.686	1.798
6	.612	.734	.857	.979	1.102	1.224	1.469	1.714	1.836	1.958

The discharge per stroke as shown by the above table may be multiplied by the number of strokes per minute to find the discharge in gallons per minute.

The power required for pumping will depend on the number of gallons per minute one wishes to pump and the total lift.

The total lift is the vertical distance from the surface of the water in the well to the highest faucet or to the storage tank plus the friction loss in the pipes. If the length of the distribution pipe is over 100 feet the loss by frictional resistance in feet of lift should be determined and included in the total lift. The following table gives the frictional loss in feet of lift per 100 feet in pipes from 3% to 4 inches in diameter, discharging from 5 to 40 gallons per minute.

FRACTIONAL LOSS IN FEET FOR 100 FEET CLEAN IRON PIPES [1]

Gallons per minute	¾ inch	1 inch	1¼ inches	1½ inches	2 inches	2½ inches	3 inches	3½ inches	4 inches
5	7.6	1.9	0.71	0.27	0.09	0.05	0.01	.....	.....
10	29.9	7.3	1.4	1.0	.28	.09	.05	0.01	.....
15	66.0	16.1	5.5	2.2	.57	.18	.09	.05	0.02
20	115.0	28.0	9.5	4.8	.96	.32	.13	.07	.03
25	179.0	43.7	14.7	6.0	1.7	.48	.23	.09	.05
30	264.0	63.2	21.0	8.6	2.1	.69	.30	.14	.07
35	372.0	85.1	28.9	11.6	2.7	.92	.39	.20	.11
40	461.0	110.0	37.0	14.9	3.7	1.2	.53	.25	.14

Ellis and Howland's experiments.

Water weighs 62.5 pounds per cubic foot and there are 7.48 gallons in a cubic foot. Divide the number of gallons pumped per minute by 7.48 to get the cubic feet pumped per minute. Multiply the cubic feet by 62.5 pounds to get the weight

of water pumped per minute. Multiply the weight by the total lift, which will give the foot-pounds of work per minute; 33,000 foot-pounds per minute equal 1 horsepower. Divide the foot-pounds per minute by 33,000 and the result will be horsepower. The horsepower as computed from the quantity pumped per minute and the total lift should be doubled, as a pumping outfit usually has an efficiency of about 50 per cent. In general, from 1 to 3 horsepower is all that is required for ordinary farm pumping. In cases where water for the house only is wanted, %4 to 34 of a horsepower is sufficient.

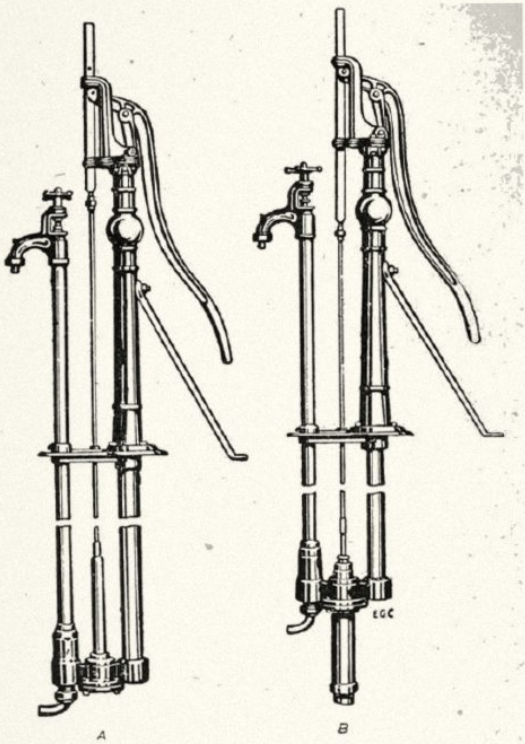
Types of Pumps.—There are several types of pumps which may be used in farm pumping. The most common are the ordinary lift pumps which simply raise the water to the ground surface from a shallow well. For elevated tank systems and pneumatic tank systems the combination lift and force pump is necessary. If a special air pump or compressor is not employed it is necessary that a combination air and water pump be used for pneumatic tanks, especially in pumping from deep wells.

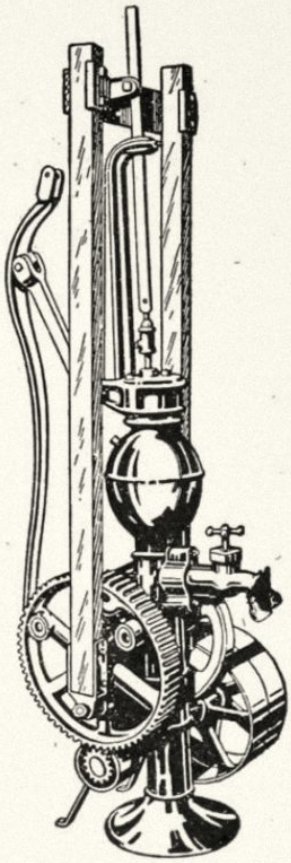
There are many types of hand force pumps for shallow and deep well pumping which may be applied to either elevated tank or pneumatic tank systems.

Figure 15 shows a deep-well windmill pumping outfit applied to a pneumatic tank system, and figure 13 shows a shallow well windmill pumping outfit supply-

ing water to a tank on the windmill tower. Figure 20 is a pumping jack which may be connected with a deep or shallow well pumping outfit applied to either system. This jack may be operated by gas engine or electric motor. Figure 21 shows two other types of windmill force pumps, A for shallow wells, and B for deep wells.

In obtaining information from the manufacturers of pumping equipment as to the particular equipment which will suit certain needs, the power required,





Fi. 20.—Pumping jack for either deep or shallow well pumping

Fic. 21. —Two types of windmill force pump: A, For shallow wells;

B, for deep wells.

etc., it is well to send data on the following: The source of water supply, whether a well, spring, or surface supply; inside diameter and total depth of the well; the distance from the ground surface to the water level in the well; the flow of the well; the number of gallons to be pumped per hour; the relative positions of the water supply and the point to which the water is to be forced; the position in which the pump is to be placed; one's preference as to pneumatic tank, elevated tank, or autopneumatic system; the kind of power to be used, and whether or not power is already available, such as electric motors, windmills or gasoline en-

gines, with a complete description of the power, its revolutions per minute, voltage, cycles, phase, direct or alternating current, etc.

Where a supply of pure water may be obtained in the immediate neighborhood, which is so situated that a considerable fall may be obtained within a reasonable distance, a hydraulic ram may be used for pumping.

Hydraulic Ram—Hints on Installation and Operation—The hydraulic ram is a simple though wasteful machine, which utilizes the momentum of a stream of water falling a small height to elevate a portion of that water to a greater height. A complete installation consists of a drive pipe, ram, and deliv-

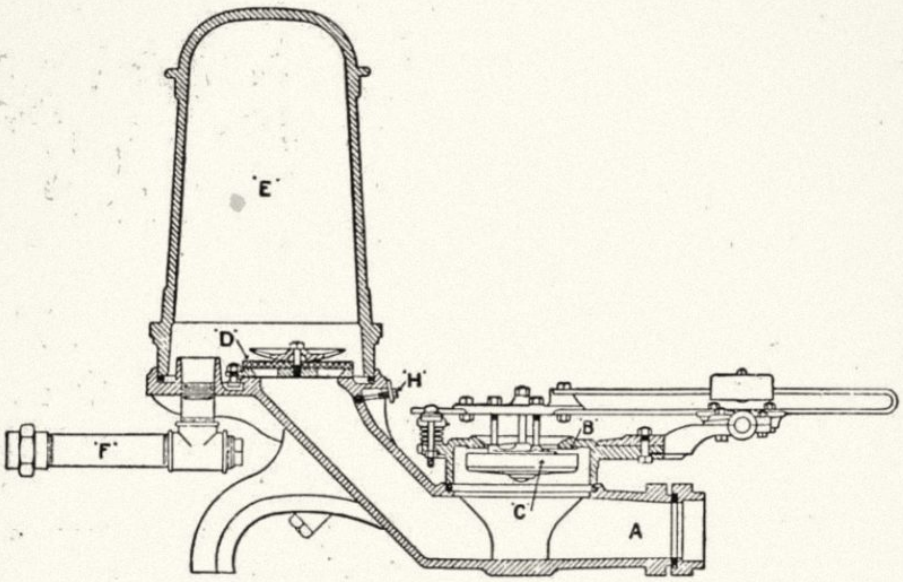


FIG. 22.—Cross section of simple type of hydraulic ram.

ery pipe, and the ram itself consists of an air chamber, dash valve, delivery valve, and body pipe.

The hydraulic ram is often used to elevate water from a pure spring. Water may be elevated from streams to stock tanks, but this water should not be used for household purposes. The flow of the spring should not be less than one-half gallon per minute. It is necessary that there be considerable difference in elevation between the level of the supply and a convenient location for the ram within a reasonable distance. Figure 22 shows a cross section of a simple type of hydraulic ram. Figure 23 shows the usual relative positions of spring, ram, and storage tank.

The operation of a ram may be briefly explained as follows: the water flowing down the drive pipe acquires a certain energy due to its weight and velocity and, upon entering the body pipe of the ram, strikes the open dash valve with considerable force, which is sufficient to close it. The result is, that the water piles up and exerts an interior pressure, which causes the delivery valve to open, admitting water into the air chamber. (See Fig. 22). The water then rebounds

back up the drive pipe until forced down again by the weight of the water in the drive pipe, and the operation is repeated. During each momentary reflow a small quantity of water is forced into the air chamber, compressing the air. The resulting air pressure forces water up the delivery pipe to the point of delivery. The operation of a ram is continuous, once started, until the valves become worn. The ordinary small ram completes its cycle about 60 times a minute, the length of stroke of the dash valve governing the number of pulsations per minute.

The length of drive pipe is most important and is governed by the ratio of the fall to the elevation. If too long or too short the automatic supply of air is interfered with and the efficiency impaired. The length of drive pipe is usually

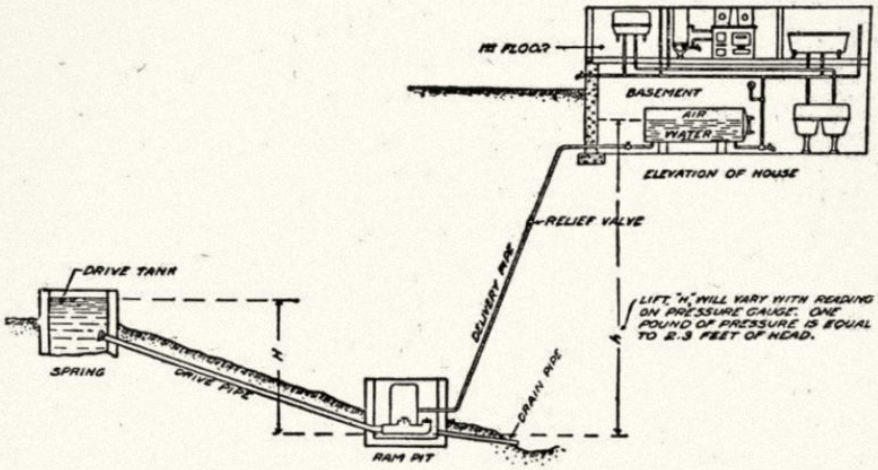


Fig. 23.—Hydraulic ram pumping to a pneumatic tank supply system, showing usual relative position

of spring, ram, and storage tank.

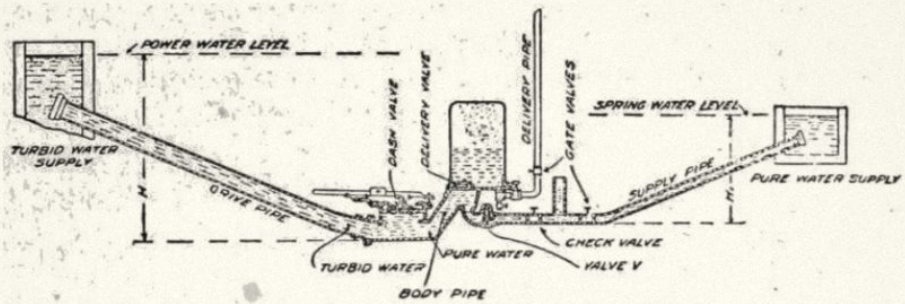
about 7 times the height of fall, although this may vary between 5 and 10, depending on the height and distance to which water is to be delivered. The diameter of the drive pipe is usually twice that of the discharge pipe.

The proper size of ram to suit certain conditions depends on the following: (1) The flow of water from the source of supply, determined by the time necessary to fill a vessel of known capacity or by weir measurement; (2) the difference between the level of the supply and the lowest point within a reasonable distance for the location of the ram; (3) the distance between the source of supply and the proposed location of the ram; (4) the difference in level between the ram location and the highest point to which water is to be delivered; (5) length of pipe necessary to conduct the water to the point of delivery. In purchasing a ram this information should be sent to manufacturers.

Sometimes a double-acting ram is installed where there is a spring too small to operate a single-acting ram but located near a brook from which an ample supply and fall can be obtained to operate the ram. These are so constructed that if properly installed under a fall of at least 2 feet below the spring and 3 feet below the brook, it is impossible to deliver anything but the spring water.

Figure 24 illustrates this method. If this method is preferred, it should be so stated in purchasing. The following table gives approximate sizes of hydraulic rams to suit certain conditions.

There are four separate problems connected with the hydraulic ram. These, with practical examples, are described by W. C. Davidson as follows: (1) Given the fall, lift, and quantity of water desired, find the necessary supply at spring.



IG. 24.—Double acting hydraulic ram, showing method of using a turbid creek supply to pump clea spring water.

(2) Given the lift, quantity of drive water, and quantity of water desired, find the fall required. (3) Given the fall, lift, and quantity of drive water, find the quantity of water supplied to the storage tank. (4) Given the fall, quantity of drive water, and quantity of water desired, find how high this water can be pumped.

SIZES OF HYDRAULIC RAMS

Number	Dimensions			Size of drive pipe	Size of delivery pipe	Quantity per minute required to operate engine	Least fall recommended
	Height	Length	Width				
	Ft. Ins.	Ft. Ins.	Ft. Ins.	Ins.	Ins.	Gallons	Feet
10.....	2 2	2 10	0 12	1¼	¾	2– 6	2
15.....	2 2	3 0	0 12	1½	¾	6–12	2
20.....	2 5	3 3	1 2	2	1	8–18	2
25.....	2 5	3 4	1 3	2½	1	12–28	2
30.....	2 7	3 7	1 3	3	1¼	20–40	2
40.....	3 7	4 9	1 8	4	2	30–75	2

The computations which follow are based upon an approximate rule, which is stated as follows: Multiply the fall in feet by the quantity of water supplied to the ram in gallons per minute, divide the product by the height the water is to be raised, and the result will be in gallons per minute. This may be expressed QxH in an equation as follows: q = (——), in which Q = supply of spring in gallons

h per minute. H = fall in feet from spring to ram, h—height of storage tank above the ram in feet, quantity of water pumped in gallons per minute. The result should be reduced by about one-third to allow for friction.

Example 1. It is desired to find the quantity of drive water in the spring necessary to raise 8 gallons per minute to a height of 60 feet, when the head of drive water on the ram is 8 feet. Substituting in the equation

$$Q = \frac{h \times q}{H} = \frac{8 \times 60}{8} = 60 \text{ gallons per minute.}$$

In this case about one-third of the result should be added to allow for friction in pipes, valves, etc., making the necessary drive water supply 80 gallons.

Example 2. It is desired to find the necessary fall from the spring to the ram in order to supply the storage tank with 2 gallons per minute, when the drive water supply in the spring is 10 gallons per minute and the height to which the water is to be pumped is 40 feet. Substituting in the equation

$$H = \frac{h \times q}{Q} = \frac{2 \times 40}{10} = 8 \text{ feet.}$$

Add one-third to allow for friction, making the total fall necessary 10.66 feet.

The following table gives commercial estimates of the quantities of water delivered in 24 hours under certain conditions:

Capacity of hydraulic rams.

Power head in feet.	Pumping head in feet—																
	4	10	15	20	30	40	50	60	70	80	90	100	120	140	160	180	200
2	540	192	128	96	64	43	29	24	.....	.....	.....	.....	.....	.....	.....	.....	.....
3	.....	301	192	144	96	72	58	43	37	27	24	.....	.....	.....	.....	.....	.....
4	.....	432	256	192	128	96	77	64	55	43	38	29	24	.....	.....	.....	.....
5	.....	540	345	240	160	120	96	80	69	60	53	43	30	26	.....	.....	.....
6	.....	.....	432	302	192	144	115	96	82	72	64	57	43	31	27	24	.....
7	.....	.....	505	378	235	168	134	112	96	84	75	67	50	36	31	28	25
8	.....	.....	.....	432	270	192	154	128	110	96	86	77	64	55	43	38	29
9	.....	.....	.....	485	300	216	173	144	124	108	96	86	72	62	54	43	39
10	.....	.....	.....	540	360	252	192	160	137	120	107	96	80	68	60	53	43
12	.....	.....	.....	.....	430	301	230	192	165	144	128	115	96	82	72	64	57
14	.....	.....	.....	.....	505	353	270	224	192	168	150	135	112	96	84	75	67
16	.....	.....	.....	.....	.....	432	323	257	220	192	171	154	128	110	96	85	77
18	.....	.....	.....	.....	.....	486	390	303	247	216	192	173	144	124	108	96	86
20	.....	.....	.....	.....	.....	540	430	336	288	240	214	192	160	137	120	107	96
22	.....	.....	.....	.....	.....	.....	475	370	303	264	235	212	176	151	132	118	105
24	.....	.....	.....	.....	.....	.....	520	405	346	288	256	230	192	164	144	128	115
26	.....	.....	.....	.....	.....	.....	.....	470	375	328	278	250	208	178	156	139	125
28	.....	.....	.....	.....	.....	.....	.....	505	430	354	300	269	224	192	168	149	134
30	.....	.....	.....	.....	.....	.....	.....	540	465	405	336	288	240	206	180	160	144

Multiply factor opposite "power head" and under "pumping head" by the number of gallons per minute used by the engine and the result will be the number of gallons delivered per day. Example: With a supply of 6 gallons per minute, 10 feet fall, 40 feet elevation, No. 10 or 15 engine will deliver 1,512 gallons per day: 6×252=1,512.

This table will give only approximate quantities since the results will vary with tho length of delivery pipe. Due consideration of pipe friction will give more correct results.

'Tho efficiency developed is governed by the ratio of fall to pumping head, being greatest for a ratio of I to 2} or 1 to 3, and the ram will not usually work well when the ratio is over 1 to 26, friction in the delivery pipe being duly considered.

Example 3. It is desired to find how much water will be delivered into the storage tank if the drive water supply is 6 gallons per minute, the fall is 10

feet, and the height to which the water is to be pumped is 40 feet. Substituting in the equation

$$q = \frac{2 \times H}{h} = \frac{6 \times 10}{40} = 1.5 \text{ gallons per minute.}$$

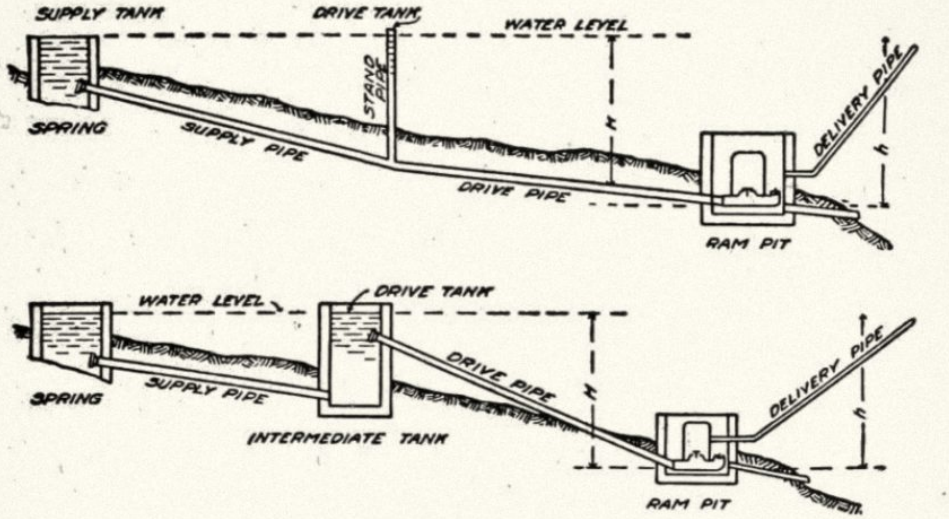
Deduct one-third of this result to allow the fraction, making the quantity delivered per minute 1 gallon.

Example 4. It is desired to find how high 1 gallon per minute can be pumped if the drive water supply is 4 gallons per minute and the fall is 15 feet. Substituting in the equation

$$h = \frac{Q \times H}{q} = \frac{4 \times 15}{1} = 60 \text{ feet.}$$

Deduct one-third to allow for friction, making the result 40 feet.

The above computations are only approximate, but should give a good general idea of the operation of a ram.



FG. 25.—Two methods of securing the necessary fall in drive pipe.

The spring should be walled in to form a reservoir as shown in the discussion of springs, page 115. If a stream is used it should be dammed back until a sufficient flow and fall is obtained.

Rams may be obtained to supply water for both elevated tank and pneumatic tank systems. In purchasing a ram this should also be specified.

In order to obtain the desired fall it is often necessary to convey the water a greater distance than the length of drive pipe used. Figure 25 illustrates two methods of securing the necessary fall.

Tt is necessary to provide a shelter for the hydraulic ram to prevent freezing in cold weather. The pipes should also be placed below the frost line. In setting a ram the foundation should be firm and level. The drive pipe should be laid on

a perfectly straight incline without bends or curves, except where the pipe enters the ram, and this should be made by bending the pipe. Fittings should not be used. The upper end of the drive pipe should be sufficiently below the surface of the water to prevent air suction—at least a foot. A good open strainer should be provided at the upper end. Above all things the drive pipe should be airtight

The delivery pipe may be laid with the necessary bends, according to the usual practice in laying water pipes, but all pipes should be connected before starting the engine and they should be left uncovered until all leaks are stopped. However, there should be as few bends and elbows in the delivery pipe as possible in order to reduce friction.

Manufacturers of hydraulic rams should supply directions for the proper installation, operation, and care of their particular rams, and these directions should be carefully followed.

Plumbing.

Important Points to be Considered——The important points to be considered in the arrangement of a plumbing system are (1) durability of material and construction, and (2) simplicity. Avoid any complication of pipes and arrange the water pipes so as to carry the water to the point of discharge in as nearly a straight line as possible. The use of lead pipe or lead-lined receptacles for drinking water should be avoided in small private systems.

Water Plumbing.—The main pipe from the supply tank should be about 1% inches in diameter and never less than 1 inch. It leads to the kitchen range and then branches. One branch conveys cold water to the fixtures and the other conveys water through the heater, through the hot-water tank, and thence to the hot-water fixtures. The hot-water pipe should parallel the coldwater pipe but should not be so close that the temperature of either will affect that of the other. The arrangment of water pipes, hot-water tank, etc., is shown in figure 14. The hot-water pipes are shown in black. All water pipes should be put in with red lead and all fittings should be screwed tight. The natural direction of travel of hot water is upward, and this should be aided, in arranging the hotwater pipes, as much as possible.

The sizes of pipes generally used for supplying water to the various fixtures are given in the following table:

SIZES OE WATER PIPES IN BUILDINGS

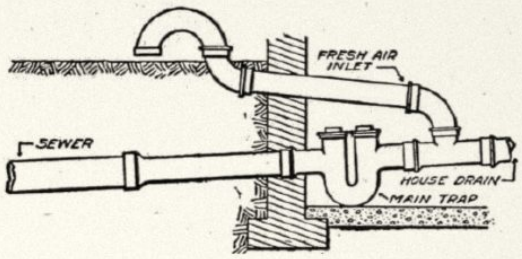
Supply Branches	Low Pressure	High Pressure	Supply Branches	Low Pressure	High Pressure
	Inches	Inches		Inches	Inches
To bath cocks	¾–1	½– ¾	To water-closet flush pipes	1¼–1½	½–⅝
To basin cocks	½	⅜– ½	To kitchen sinks	⅝– ¾	.....
To water-closet flush tank	½	½	To pantry sinks	½	⅜–½
To water-closet flush valve	1 –1¼	¾–1	To slop sinks	⅝– ¾	½–⅝

. WATER SUPPLIES, PLUMBING AND SEWAGE 135

All water pipes should have sufficient slant to drain them back into the tank or drainage system. A drain pipe and cock should be provided at the low point in the system, so that in extremely cold weather the system may be drained into the sewer or drainage system to prevent freezing. This necessitates a stop cock on the pressure-tank outlet to prevent draining the tank.

Pipes should be kept from the outer walls to prevent freezing. Also pipes located where they are in any danger whatever of freezing should be boxed in sawdust or some other nonconducting material.

Since a plentiful supply of hot water is convenient and a large quantity retains heat for some time, it is well to provide a fairly large hot-water tank. However, the size of boiler depends on the existing conditions, such as the water supply and the size of building. A safe rule is to allow a 35 or 40 gallon boiler



Fic. 25,—Sewer trap at house foundation, showing ventilator

to a building having one bath-room and to add 30 gallons additional capacity for every extra bath-room. One hundred square inches of water-back heating surface is sufficient for a 40-gallon boiler.

Boilers should be galvanized inside and out, particularly inside. Copper boilers are preferable if properly coated inside with block tin. These are classed as light, heavy, and extra heavy, the latter being tested to 150 pounds water pressure. Ordinary steel or iron boilers are tested to 150 pounds water pressure and extra heavy ones to 250 pounds pressure. The latter should be used when the gauge pressure is more than 40 pounds per square inch. The following table gives standard sizes of galvanized boilers:

STANDARD SIZES OF GALVANIZED BOILERS

Capacity	Length	Diam.	Capacity	Length	Diam.	Capacity	Length	Diam.
Gallons	Feet	Inches	Gallons	Feet	Inches	Gallons	Feet	Inches
18	3	12	36	4½	14	82	5	20
21	3½	12	40	5	14	98	6	20
24	4	12	42	4	16	100	5	22
24	3	14	47	4½	16	120	6	22
27	4½	12	48	6	14	120	5	24
28	3½	14	52	5	16	144	6	24
30	5	12	53	4	18	168	7	24
32	4	14	63	6	16	182	8	24
35	5	13	66	5	18			
36	6	12	79	6	18			

Sewer Plumbing.—The sewer plumbing serves as a drain for the water plumbing. The drainage system should be so constructed as to carry away completely everything emptied into it, and it should be constantly vented, frequently and thoroughly flushed, and have each of its openings into the house securely guarded. All drains, soil pipe, and waste pipe should be water-tight and airtight. The soil pipe or house-drainage main begins at the sewer opening and passes up through the house as nearly vertical as possible and out through the roof for free ventilation. It should be at least 4 inches in diameter, of extra heavy cast iron, and all joints should be tightly calked with lead and oakum. All discharge from the wash basins, sinks, and toilets empties into the soil pipe, and connections should be tightly made. The sewer inside the cellar wall should always be soil pipe; tile should never be used except outside of the wall. A soilpipe trap should be provided at the house foiindation, as shown in Figure 26. Every fixture should have a trap to prevent foul air from coming back through the waste pipe. Vent pipes should be provided on all waste pipes to prevent siphonage and the consequent destroying of the traps. Figure 2% shows a good arrangement of sewer plumbing. Note the traps and vent pipes on each waste pipe. The least sizes of waste and vent pipes are given in the table below.

SIZES OF WASTE AND VENT PIPES

Name of Pipe	Diameter	Name of Pipe	Diameter
	Inches		Inches
Main and branch soil pipe	4	Wash tubs, 1½-inch waste pipe to 2-inch trap for 2 tubs	1½–2
Main waste pipe	2	Waste pipes for 3 or 4 tubs	2
Branch waste pipes for kitchen sinks	2	Main vents and long branches	2
Bath or sink waste pipes	1½–2	Branch vents for traps over 2 inches	2
Basin waste pipe	1¼–1½	Branch vents for traps less than 2 inches	1½
Pantry sink waste pipe	1½		
Water-closet trap	3½–4		

All plumbing should be tested by filling with water or smoke to detect leaks.

Sewage Purification and Disposal.

The problem of the purification and disposal of farm sewage by small private systems, differs somewhat from that of city sewage disposal, owing principally to the extreme fluctuations in flow, small size of the system, and variation in the quality of the sewage.

The process of sewage disposal is partly mechanical and partly bacterial, consisting of (1) preliminary or tank treatment and of (2) final treatment, which is application to a natural soil by surface or subsurface distribution or to an especially prepared filter.

Preliminary or Septic-Tank Treatment.—The exact nature of the action which takes place in a septic tank is a subject of dispute among sanitary experts and bacteriologists. Several theories have been advanced, but it is apparent that no definite conclusion has been reached. Some authorities advocate the use of. open ventilated tanks, others advocate the use of air-tight tanks.

Experience has shown that, in a small sewage disposal system, a dark, airtight tank of sufficient capacity and so constructed that sewage may remain in it entirely at rest for a period of from 18 to 24 hours gives the best results and the least annoyance. The solid matter settles out in such a tank and is partially liquefied, deodorized, and destroyed by countless numbers of bacteria, which

thrive in filth and live without air. Some authorities assert that these bacteria also slightly affect the dissolved organic matter in raw liquid sewage.

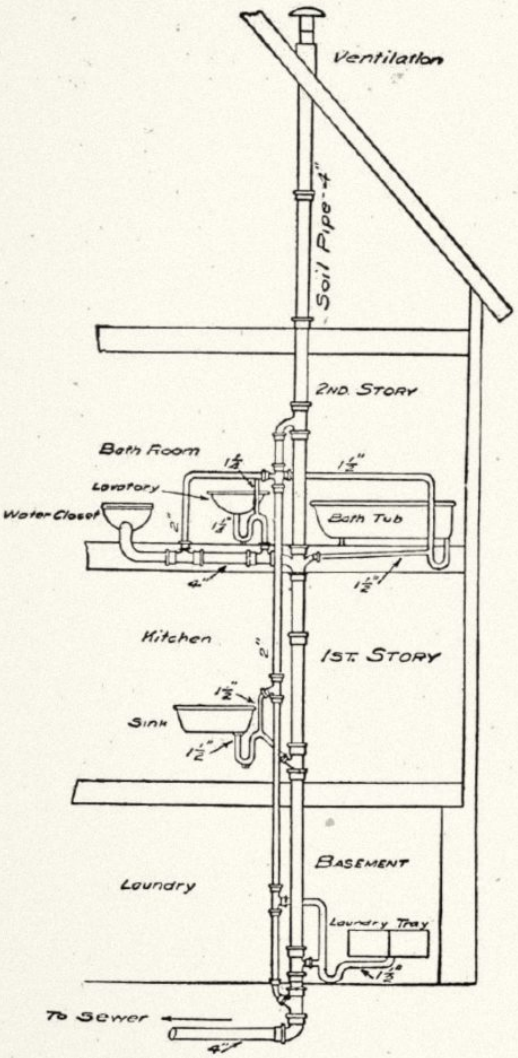


FIG. 27.—Plumbing system for sewage disposal.

In such a tank a thick scum forms on the surface of the sewage, which protects the bacteria from the incoming air and is evidence of good bacterial action. The breaking up or disturbance of this scum will destroy the bacterial action for the time being and is likely to, cause considerable annoyance of bad odors.

Final Treatment.—It is found that the septic tank effects only about 40 percent purification. The liquefying action in the tank, however, makes it possible to subject the sewage to a final treatment by filtration or distribution in a natural soil. This final purification is effected by means of bacteria which work in the air. Therefore it is necessary that the sewage be applied to the final disposal system in intermittent doses so that the system may have a chance to air out. If the sewage is applied continuously and in such quantities that 'the system is kept saturated, the filter or disposal area becomes water-logged and “sewage sick” and ceases to be effective. It is therefore necessary

that the final treatment system be of sufficient capacity to dispose of each sewage quickly.

Double-Chamber Septic-Tank Systems.—The septic tank for a small sewage-disposal system should ordinarily consist of two chambers. In this type of tank the sewage is received, settled, and partially purified in one chamber and collected and discharged from a second chamber. This type of tank, if properly designed, should give satisfactory operation, since the sewage in the settling chamber suffers little disturbance, and the discharge to the final disposal system may be made intermittent by means of an automatic siphon placed in the discharge chamber.

Design.—Practice indicates that the settling chamber of a small septic tank should have a capacity of from 5 to 15 cubic feet or from 40 to 80 gallons per person in the family. Some allow an average of 10 or 11 cubic feet per person. The best results are obtained when the capacity approaches the larger limit, so that 18 to 36 hours' sewage from the house may be held at one time, thus causing the sewage to remain in the tank and undergo sedimentation and bacterial action for this length of time. But care should be taken not to make the tank so large that liquefied sewage will remain in it more than 36 hours, for in that event putrefaction is likely to set in. For this reason one should make an accurate estimate of the daily sewage flow, which will be practically equal to the daily water consumption' Although a depth of 3 feet may be sufficient for some classes of sewage, it is better to have the depth from 4 to 8 feet, according to the number of people, in order to give the sludge a good chance to settle and liquefy. The width of the chamber may ordinarily be about one-third or onéhalf the length, although this may vary for economy and convenience. However, the width should not be less than 3 feet.

The inlet from the house should be provided with an elbow, so that the discharge will be at least a foot below the contained sewage, thus preventing disturbance of the surface scum. The outlet from the settling chamber should be equipped in the same way. Where the entrance and discharge velocities are very strong, baffle walls of wood or concrete should be placed before these openings to break the current. These precautions are especially beneficial in the smaller sized tanks.

The discharge chamber should be of such capacity and depth as to discharge about every 10 to 12 hours. It may be desirable to discharge at more or less frequent intervals, according to the nature of the soil in the disposal area, and this may be controlled by the arrangement of the discharge chamber and siphon. Where little outlet fall is available it is possible to so construct the discharge chamber that its floor will be considerably above that of the settling chamber.

The capacity and depth of discharge chamber and the size of siphon will depend on the number of persons served and the means of disposal. If a sand filter is used or a distribution system in heavy loam, the discharge chamber must be larger and deeper, in order that the discharge interval may be lengthened and the distribution system be given ample time to aerate. If the distribution is in sandy or very porous soil the discharge may be more frequent.

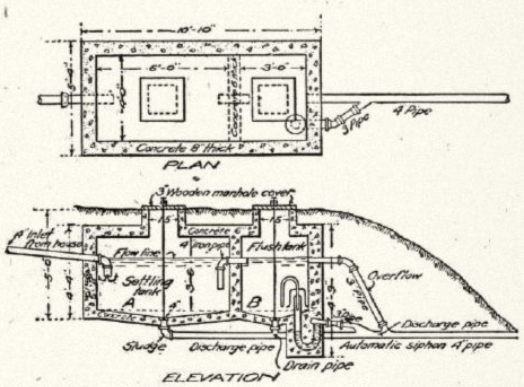
The following table of dimensions of septic tanks, suggests sizes of settling and discharge chambers and the corresponding siphon sizes applying-to various average conditions. The depths of siphon chamber given are the minimum allowable.

DIMENSIONS OF SEPTIC TANKS

Number Persons	Settling Chamber			Siphon Chamber						Siphon Diameter
				Sand Filter or Heavy Loam Distribution			Sandy or Porous Soil Distribution			
	Width Inside	Length Inside	Depth	Width Inside	Length Inside	Minimum Depth	Width Inside	Length Inside	Minimum Depth	
	Feet	Feet	Feet	Feet	Feet	Ft. In.	Feet	Feet	Ft. In.	Inches
6	4	6	3½	4	3	2 4	3	2	2 4	3
8	4	6½	4	4	4	2 4	3	2½	2 4	3
12	4	7	5	4	5	2 5	3	4	2 5	4
15	4	8	5	4	6	2 5	3	4	2 5	4
25	4	10	5	4	6½	3 2	3½	4	3 2	5
35	4½	12	5	4	6½	3 2	3⅔	4½	3 2	5

The above table is computed on the basis that the inlet and outlet of the settling chamber should be placed with their inverts 12 inches below the roof of the tank, thus making the depth of sewage in both settling and discharge chamber 12 inches less than the mean inside depth.

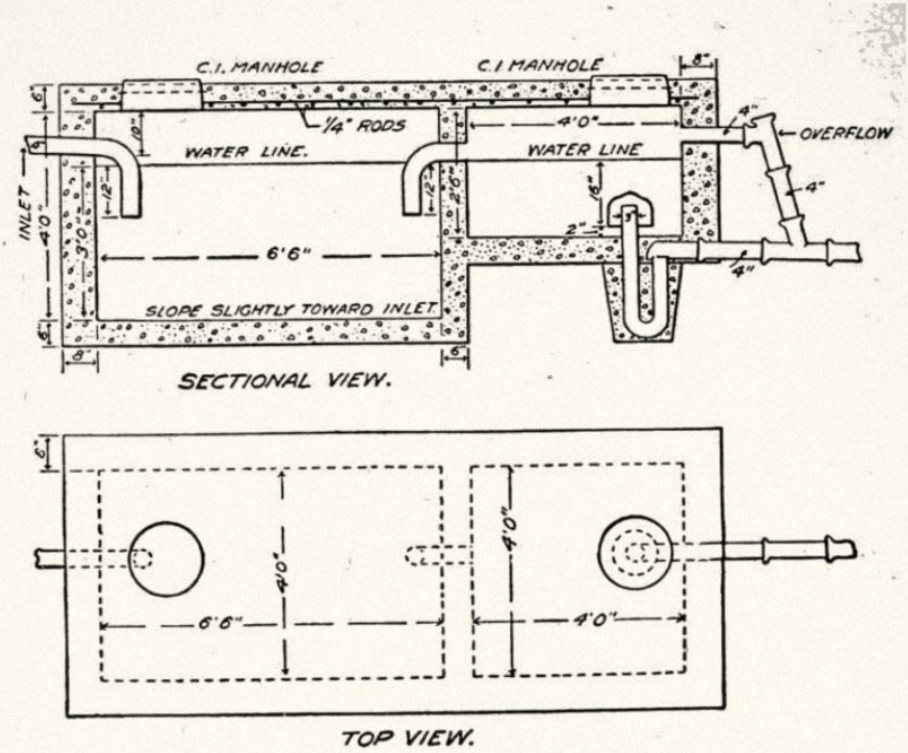
The tank dimensions given in the above table, it should be remembered, are for average cases only and are not standard for all such cases, They are subject to such variations to suit local conditions as the farmer's judgment indicates;



Double chamber septic tank for six people, suited to conditions where plenty of ou available.

yet care should be taken not to vary any of the essential dimensions and not to go below the given minimum depth of siphon chamber.

Figure 28 shows a type of double chamber septic tank for a family of six people, designed by W. C. Davidson. Figure 29 is another type of septic tank for a family of eight people. These tanks are suited to conditions where plenty



Fic. 29.—Double chamber septic tank for eight people.

of outlet fall is available. Figure 30 is a double-chamber tank for a family of six persons, designed by C. A. Ocock of the Wisconsin Agricultural Experiment Station. This tank is suited to flat ground where outlet fall is difficult to obtain, as will be noted by the difference in elevation between the floors of the two chambers. For satisfactory operation a small septic tank should not be built of smaller size than for six persons.

Location and Construction——The septic tank, although air-tight and supposedly water-tight, should be located as far from the house and the well or spring as convenience and local surroundings will permit. It thus reduces the danger of pollution or nuisance in case of leakage or improper operation of the system.

The sewer from the honse should be of vitrified sewer pipe, usually of 4 inch size, with tightly cemented joints, and should be laid to a grade not less than

9 inches per 100 feet. Where the fall from the house to the tank is excessive, it is a good idea to lay the last 100 feet of tile at the minimum grade to break up entrance velocity.

The septic tank should be constructed as nearly water-tight as possible, preferably of concrete. The walls should be 6 or 8 inches thick, the floor 4 to 6 inches thick, and the roof. about 6 inches thick and re-enforced. Some means should be provided at the bottom to aid the cleaning out of settled sludge. Either the

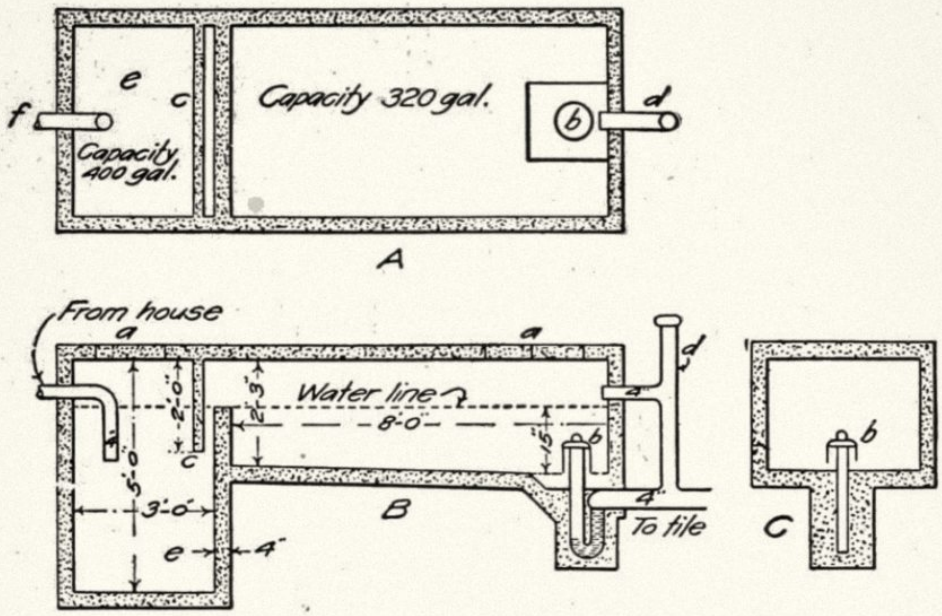


Fig. 30.—Double chamber septic tank for family of six people, suitable to conditions where outlet fall is

difficult to obtain.

floor may be sloped toward the inlet end for this purpose or a pipe with a valve may be installed below the tank, as shown respectively in figures 29 and 28. The discharge chamber should be fitted with an outlet set above the siphon which will allow the sewage to escape in case the siphon becomes clogged.

A concrete mixture of 1 part cement to 2 or 2% parts sand and 4 or 5 parts of broken stone or gravel should be used in the construction of the tank.

The Automatic Siphon.—The automatic siphon may be installed to operate as frequently as may seem desirable' Figure 31 shows a 3 or 4 inch automatic siphon installed. The siphon operates as follows: As the liquid enters the discharge chamber its weight increases with increasing depth, and the air between the water surface in the bell and the water inside the "siphon leg" is compressed. As the water outside increases in depth the compression inside becomes greater until the water outside reaches the drawing or discharge depth for the siphon. Then the inside pressure is sufficient to force the water in the

siphon leg around the bend, instantly relieving the compression. The water from the tank then rushes in to fill up the space which was occupied by the air and starts the siphon, which continues until the outside and inside pressures are again equalized.

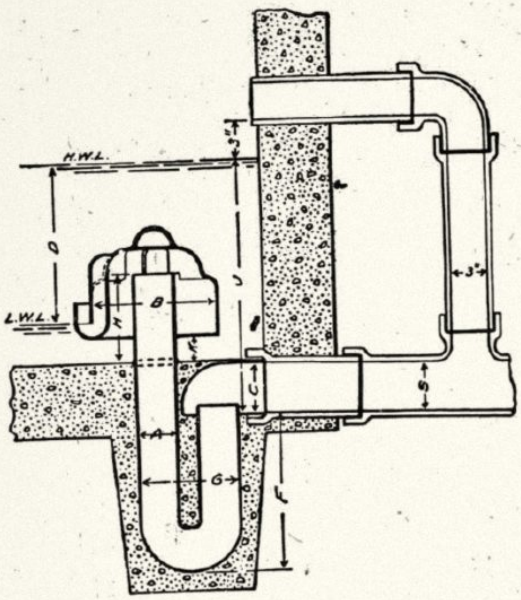


Fig. 31.—Three-inch or four-inch automatic siphon installed.

The following table gives working data and dimensions, as shown in figure 31, to be used in installing 38, 4, and 5 inch siphons. Sizes of 5 inches and over are constructed a little differently from the 3 and 4 inch sizes, although the working principles are the same.

DIMENSIONS FOR AUTOMATIC SIPHONS

Dimension	Reference Letter	Inches	Inches	Inches
Diameter of siphon	A	3	4	5
Diameter of bell	B	10	12	15
Diameter of discharge head	C	4	4	6
Drawing depth	D	13	14	23
Depth of trap	F	12	13	22
Width of trap	G	10	12	14
Height above floor	H	7¼	8¾	9½
Clearance under bell	K	2	2	3
Diameter of carrier	S	4	4– 6	6– 8
Invert to discharge line	J	20½	22¾	33½
Average discharge per second.....cubic feet..	.....	0.16	0.35	0.73

The Final Disposal System.

Disposal by Surface or Subsurface Distribution —Where the soil is porous or sandy and there is plenty of area available, which is used for no other purpose, the sewage from the septic tank may be discharged through 4inch distributiontile laid on the surface of the ground in gridiron or herring-bone fashion. Four hundred and fifty to 500 square feet of area are necessary for each person served

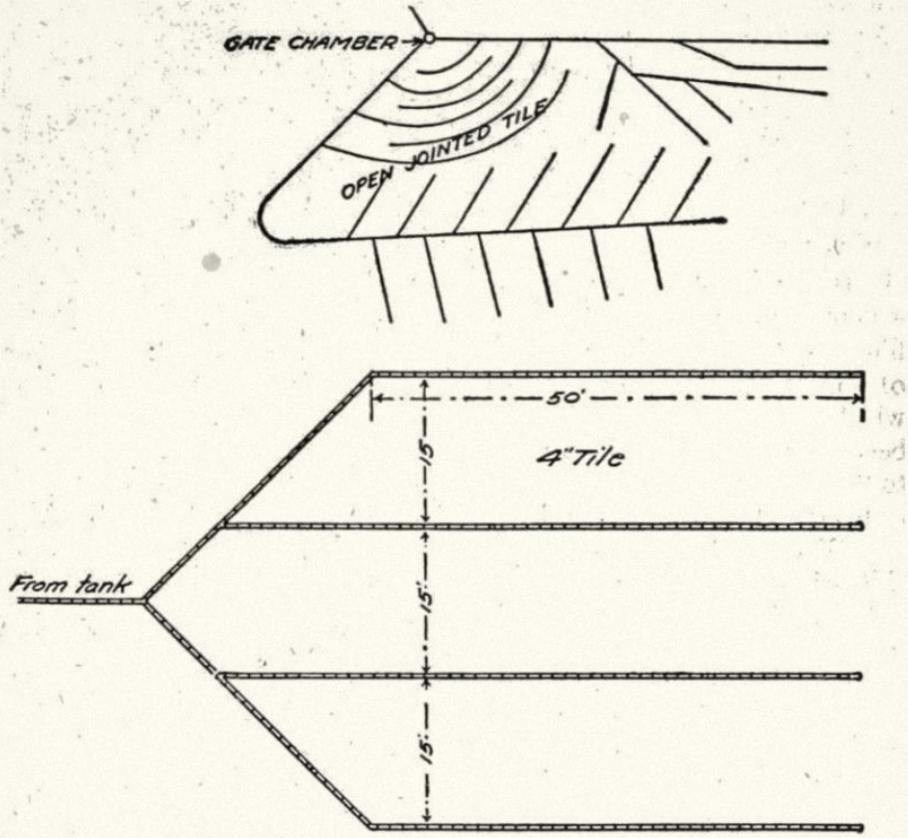


FIG. 32.—Ground plans of tile sewage disposal systems.

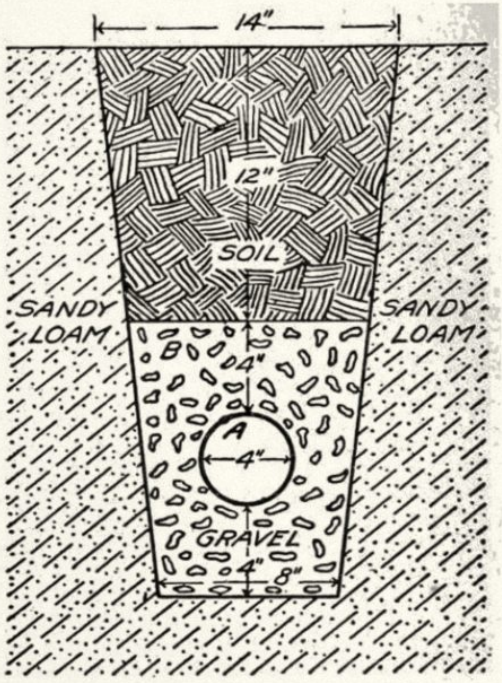
if the soil is very porous and sandy, and the soil should either be tile-drained or have natural under-drainage

A better method of disposal is by subsurface distribution. In this method the tile are placed in the ground in herringbone or gridiron fashion, not deeper than 14 or 16 inches from the surface of the soil to the top of the tile. Figure 32 shows ground plans of such systems. In very porous or sandy soils 1 foot of 4-inch tile per gallon of discharge per day is sufficient. In the heavier loam soils 2 feet of 4-inch tile are necessary and sometimes more for every gallon. A rough estimate should be made of the number of gallons of sewage in each discharge from the tank and the number of dis-charges per day. Not less than 35 feet of 4-inch tile per person should be used in sandy or porous soil and not

less than 60 feet per person in very heavy loams. In average loams 300 to 400 feet of tile are sufficient for a family of six or eight:

Aeration of heavy soils can be brought about by the use of coarse cinders or gravel laid in 12-inch to 16-inch layers in the bottom of the tile ditch with the top about 12 inches below the surface. The tile are laid in these at the usual depth. Figure 33 shows such an arrangement.

The disposal tile should have a fall not to exceed 1 inch in 50 feet, else the water will rush to the lower end and water-log the soil there. The tile are usually laid about inch and in rows about 15 feet apart. The latter distance, however, will vary with the porosity of the soil. Where there is no subsurface drainage, artificial drainage should be provided by means of tile drains laid below the sewage tile as shown in figure 34. In some cases an impervious stratum underlying the filter earth is underlain by a stratum of sand. Cases have been noted in which this impervious stratum has been broken by dynamite at 15-foot to 20-foot intervals along the tile line, thus providing natural drainage.



Disposal by Intermittent Sand Filtration.—If{ subsurface disposal is not feasible, as when the soil-is compact and nearly impenetrable or is swampy and under-drainage difficult to obtain, disposal by intermittent sand filtration is necessary.

Fic. 33.—Cross section of single tile sewage disposal The sand filter usually is a bed system, showing method of aerating heavy soils,

of sand 3 to 4 feet thick which is fine on top and gradually increase in size to coarse gravel at the bottom. The sewage from the tank is distributed over the filter by means of tile laid loose jointed over the surface in much the same manner as in the ground-surface distribution system. The filter should be sufficiently porous. There should be sufficient natural or artificial under-drainage to allow all sewage to sink away rapidly. Sewage should not stand on the surface of the filter for any length of time, as this will soon destroy its purifying properties. About 45 square feet of filter should be provided for each person served by the sewer. The area should be divided into from three to five beds so that each bed may be allowed to rest occasionally. Figure 35 shows a plan and partial section of a sand filter for a family of eight persons.

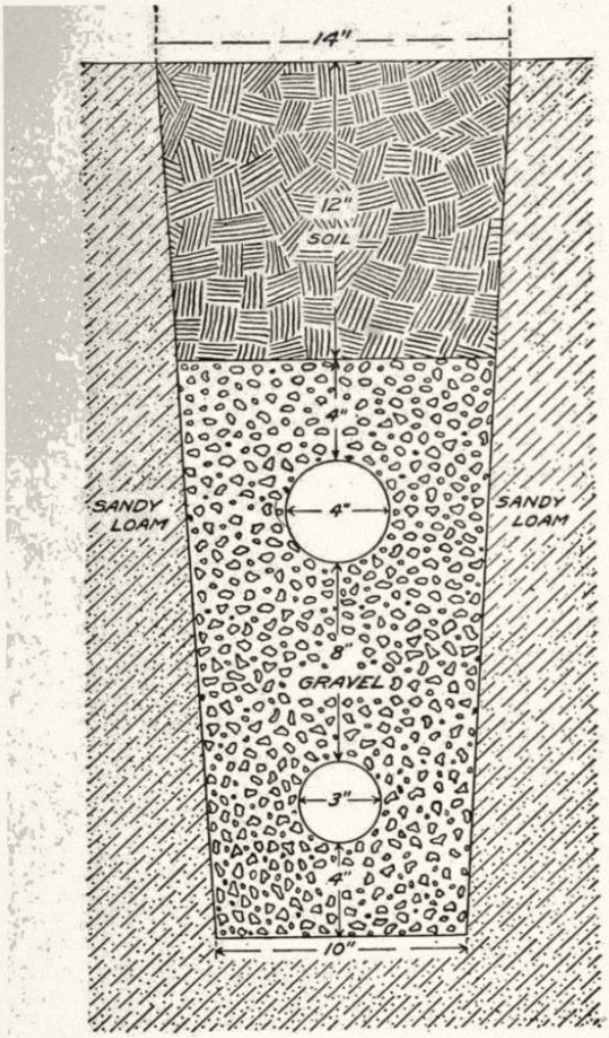
In constructing a filter a sufficient area should be leveled off and small earth embankments be made 18 inches to 2 feet high to inclose the beds. The

depth of the filtering material will depend largely on the porosity of the spbsurface and the means of under-drainage, but it is well to have it not less than 24% feet; 3 to 5 feet is better, and the depth should not exceed 6 feet. A good plan is

to allow a minimum of 1 cubic

yard of filtering material for every

50 gallons of sewage flow.



Fic. 34.—Cross section of single tile sewage disposal system, chamber es soles sey wits

Siete cecunal tlle belowetoceradediatiara!: This tank has a continuous dis-

charge, and it is necessary to use

a switch or diverting gate, as shown in figure 37, so that the liquid sewage may be intermittently diverted from one part of the disposal system to another.

The disposal system should be divided into at least two divisions for intermittent application, and the capacity of the system should be 10 to 15 per cent larger than for a double-chamber tank system.

Single-Chamber Tank Systems.

Single - chamber septic - tank systems may be made to give good satisfaction if properly designed and operated. In such a system the sewage is received, settled, partially purified, and discharged by one chamber.

There is necessarily considerable disturbance of the sewage in the tank, and, in addition, the discharge is continuous. This makes necessary two disposal systems, with a diverting gate to allow an occasional breathing spell for each system. If such an arrangement is not used the disposal system must be of much larger capacity than for the double-chamber tank system. This is necessary to prevent the continuous discharge from waterlogging the system.

If a single-chamber tank is used it should be designed and constructed on the same basis as the settling chamber of a double-chamber tank, with the elbows at inlet and outlet and baffle boards before these openings to break up the current.

Figure 36 shows a single-

The single chamber septic system requires considerable attention, since there is no provision for automatic discharge. Figures taken from the work of the Wisconsin Agricultural Experiment Station show that in the long run there is little difference in the cost of the single- and double-chamber tank system.

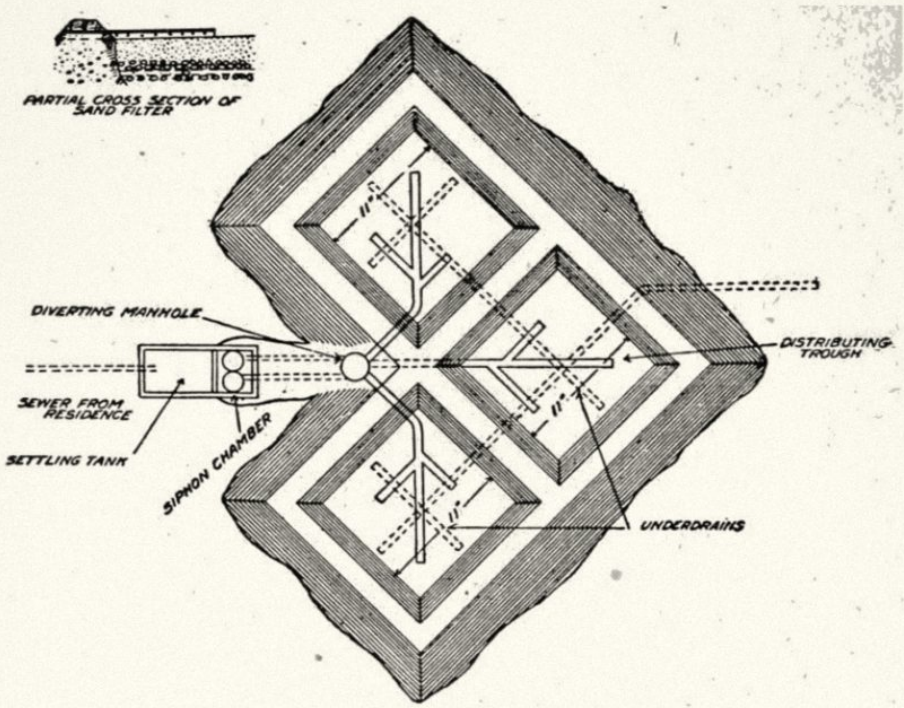
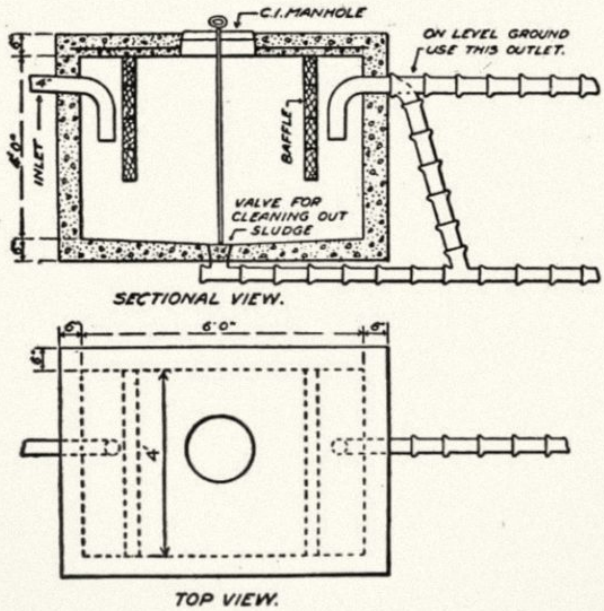


FIG. 35.—Sand filter for eight people.



Fic. 36,—Single-chamber septie tank for six people,